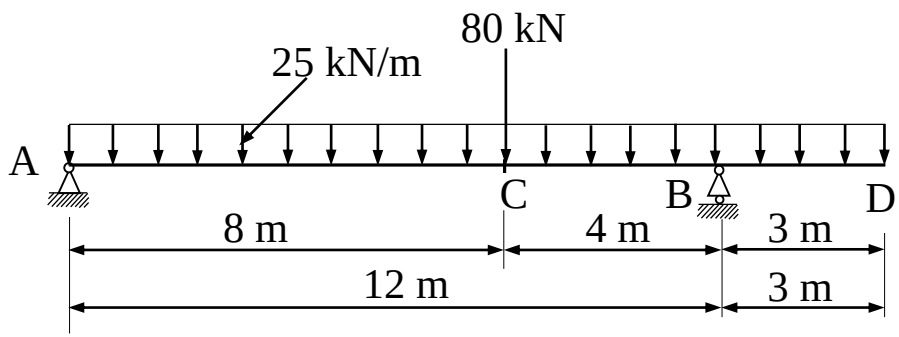
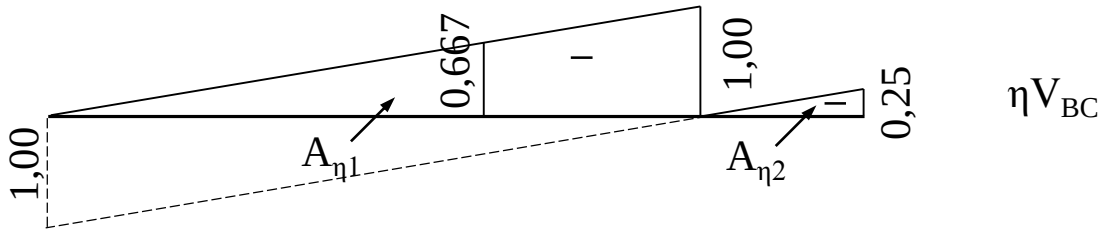
Example 1:

Estimate V_{BC}, V_{CA}, V_{CB} shear forces and M_B, M_C bending moments by using influence line.



Solution:

a) V_{BC} shear force



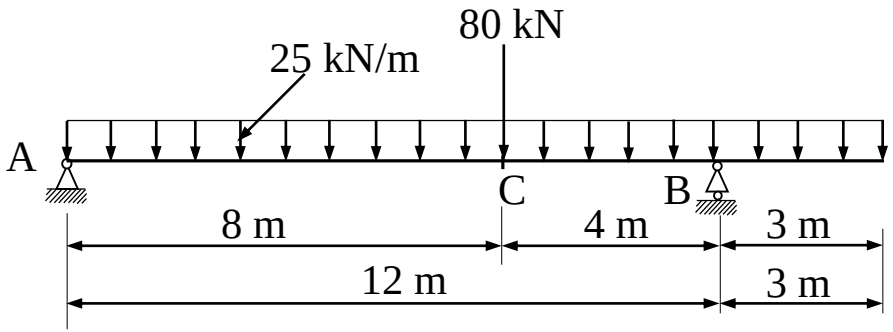
ηV_{BC}

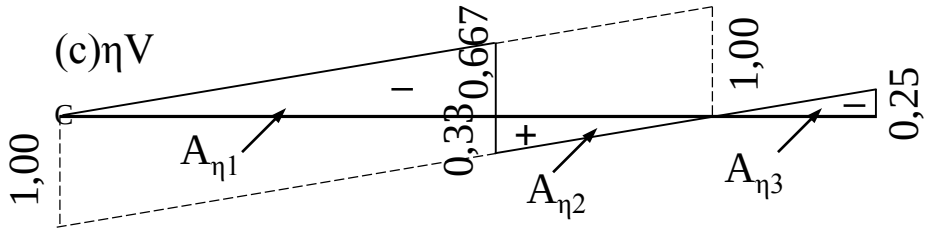
$$A_{\eta 1} = 0{,}5 \times 1{,}00 \times 12{,}0 = 6{,}00 \text{ m} \qquad A_{\eta 2} = 0{,}5 \times 0{,}25 \times 3{,}0 = 0{,}375 \text{ m}$$

$$\eta_C = 1{,}00 \times 8{,}0 / 12{,}0 = 0{,}667$$

$$V_{BC} = 25 \cdot (-6{,}0 - 0{,}375) + 80(-0{,}667) = -212{,}735 \text{ kN}$$

b) Shear force at C





Shear force at the right of the section $\eta_{CB} = -1{,}00 \times 8{,}0 / 12{,}0 = -0{,}667$

Shear force at the left of the section $\eta_{CA} = 1{,}00 \times 4{,}0 / 12{,}0 = 0{,}333$

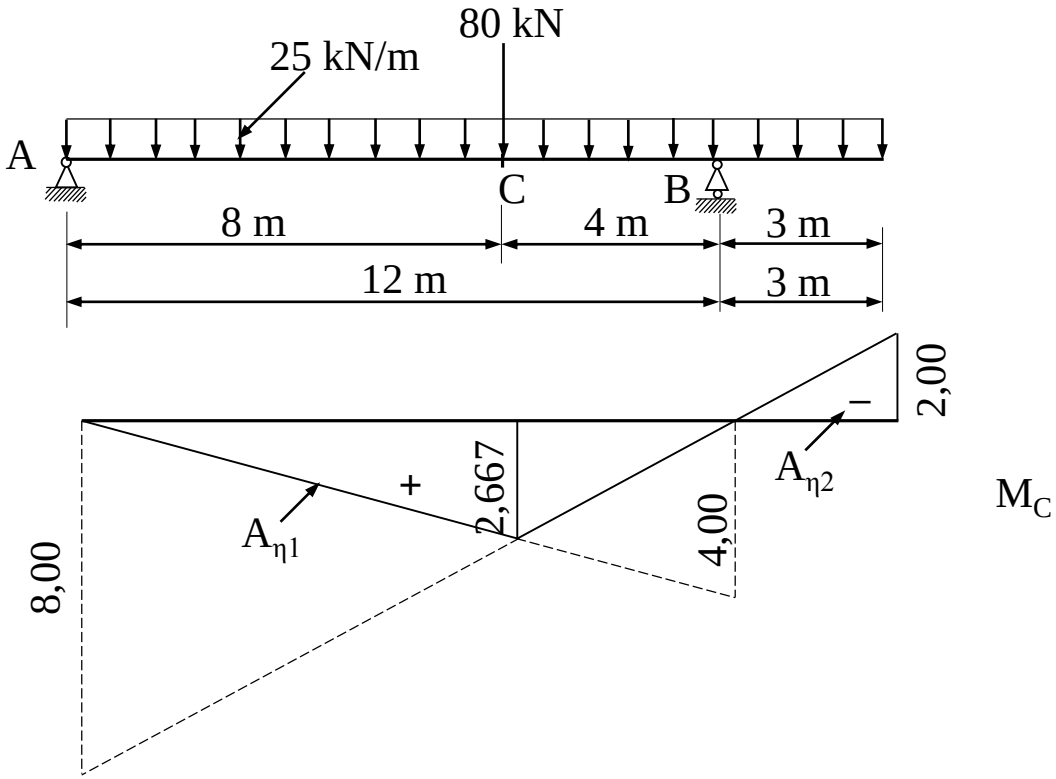
$A_{\eta 1} = 0{,}5 \times 0{,}667 \times 8{,}0 = 2{,}668 \text{ m}$ $A_{\eta 2} = 0{,}5 \times 0{,}333 \times 4{,}0 = 0{,}666 \text{ m}$

$A_{\eta 3} = 0{,}5 \times 0{,}25 \times 3{,}0 = 0{,}375 \text{ m}$

$V_{CA} = 25 \cdot (-2{,}668 + 0{,}666 - 0{,}375) + 80 \times 0{,}333 = 32{,}760 \text{ kN}$

$V_{CB} = 25 \cdot (-2{,}668 + 0{,}666 - 0{,}375) - 80 \times 0{,}667 = -112{,}76 \text{ kN}$

c) Bending moment at C

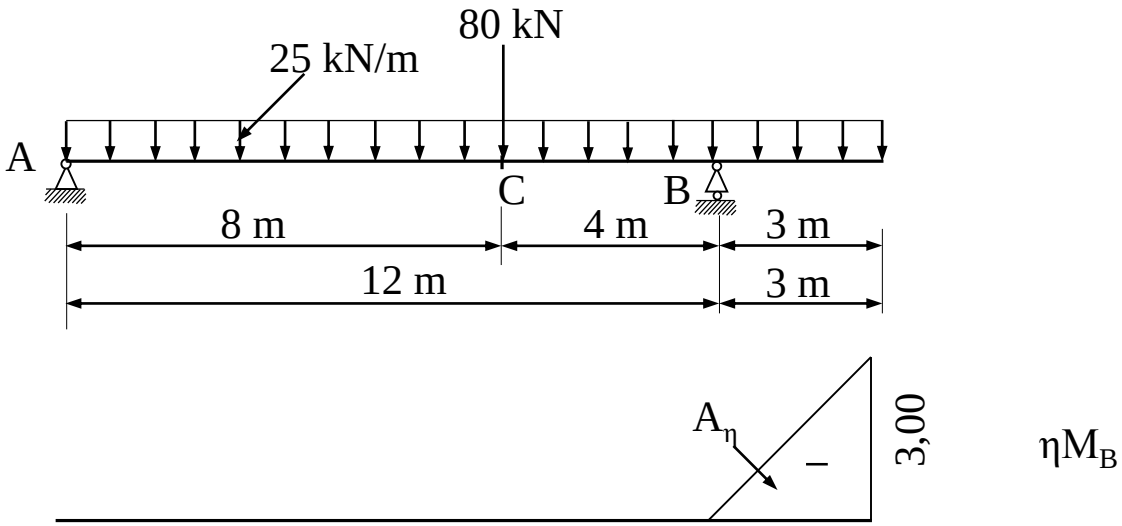


$$\eta_C = 4{,}0 \times 8{,}0 / 12 = 2{,}667 \text{ m}$$
$$\eta_D = 8{,}0 \times 3{,}0 / 12 = 2{,}00 \text{ m}$$

$$A_{\eta 1} = 0{,}5 \times 2{,}667 \times 12{,}0 = 16{,}0 \text{ m}^2 \qquad A_{\eta 2} = 0{,}5 \times 2{,}00 \times 3{,}0 = 3{,}00 \text{ m}^2$$

$$M_C = 25 \cdot (16{,}0 - 3) + 80 \times 2{,}667 \qquad M_C = 538{,}36 \text{ kNm}$$

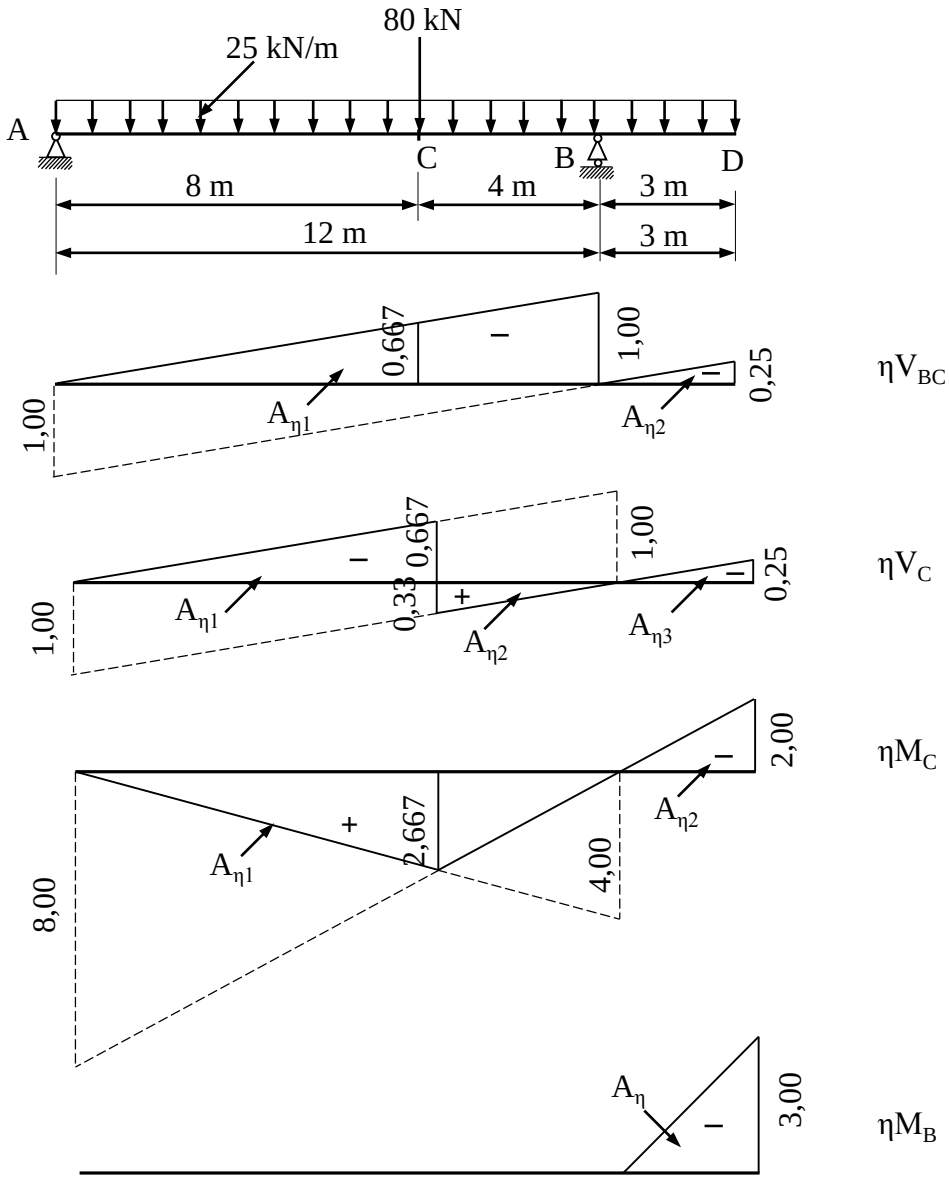
d) Bending moment at B



$$\eta_D = 3{,}0 \ \text{m}$$

$$A_\eta = 0{,}5 \times 3{,}0 \times 3{,}0 = 4{,}5 \ \text{m}^2$$

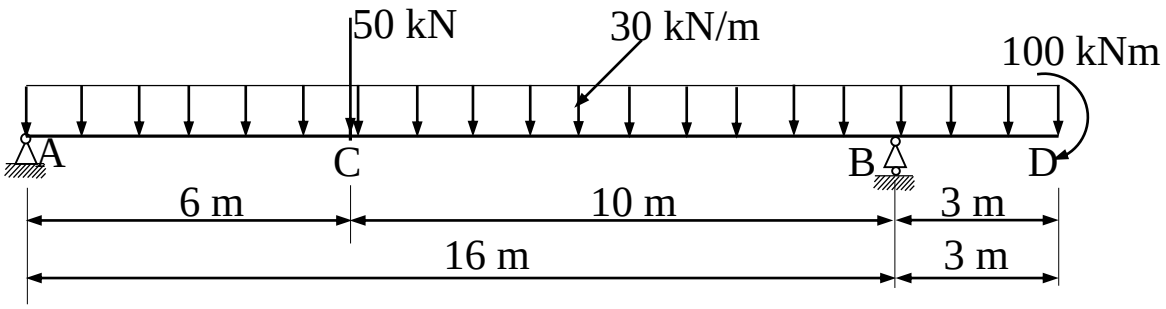
$$M_B = 25 \cdot (-4{,}5) = -112{,}50 \ \text{kNm}$$


80 kN
25 kN/m
A
C
B
D
8 m
4 m
3 m
12 m
3 m
0,667
1,00
0,25
1,00
$A_{\eta 1}$
$A_{\eta 2}$
ηV_{BC}
0,667
1,00
0,25
0,33
1,00
$A_{\eta 1}$
$A_{\eta 2}$
$A_{\eta 3}$
ηV_C
2,00
2,667
4,00
8,00
$A_{\eta 1}$
$A_{\eta 2}$
ηM_C
3,00
A_{η}
ηM_B

Example 2:

1) Estimate R_{Ay} support reaction, shear forces and bending moments at C and B sections for the given system using influence line.

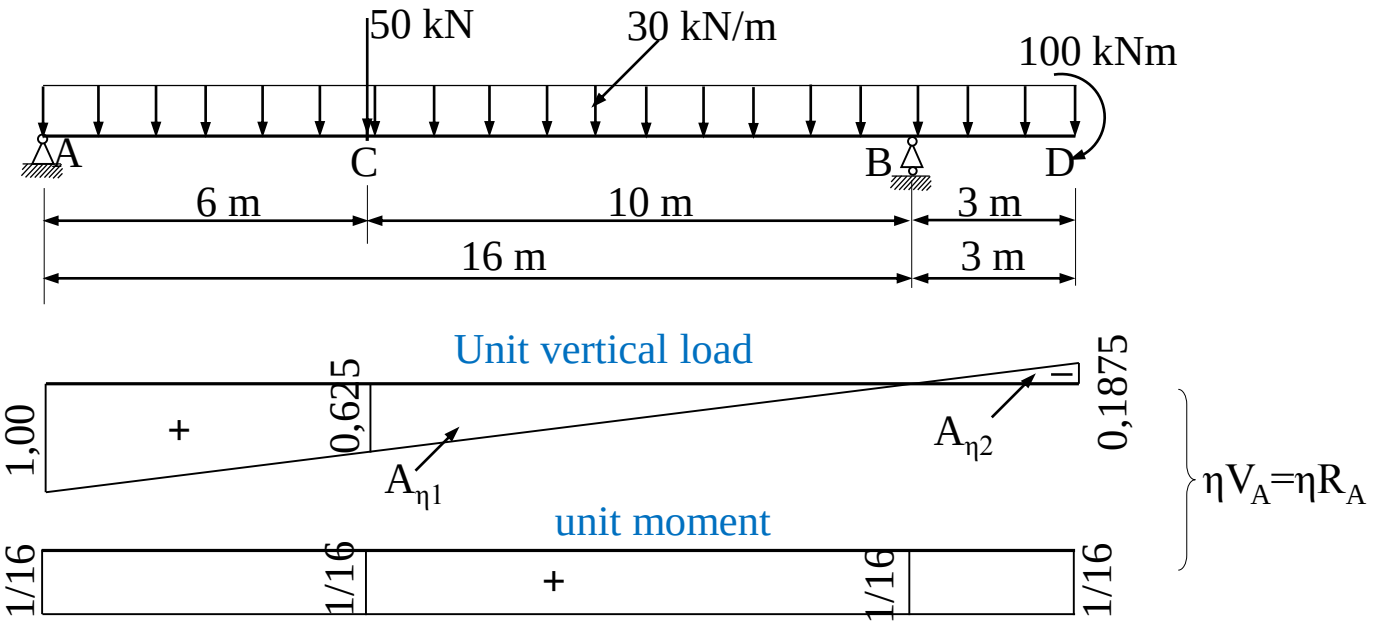
2) Estimate the maximum and minimum value of bending moment at section C due to given dead loads, 10 kN/m live uniform distributed load and 100 kN live point load.



Solution:

Since there are both vertical loads and point moment, influence lines are constructed for the unit point load and unit point moment.

1.a) R_A support reaction



For unit vertical load

$$\eta_A = 1 \qquad \eta_C = 10/16 = 0{,}625 \qquad \eta_B = 0 \qquad \eta_D = -3/16 = -0{,}1875$$

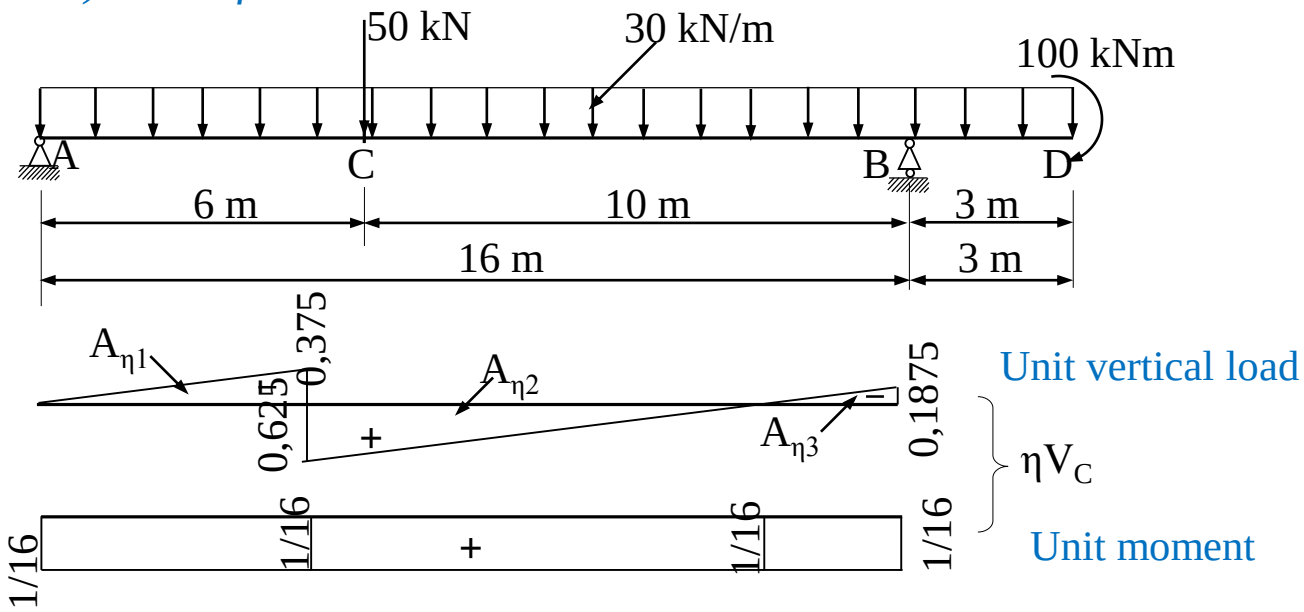
$$A_{\eta 1} = 0{,}5 \times 1{,}0 \times 16{,}0 = 8{,}00 \text{ m} \qquad A_{\eta 2} = 0{,}5 \times 0{,}1875 \times 3{,}0 = 0{,}281 \text{ m}$$

For unit moment

$$\eta_A = \eta_D = \eta_B = \eta_D = 1/16 = 0{,}0625 \text{ m}^{-1}$$

$$R_A = 30 \cdot (8{,}00 - 0{,}281) + 50 \times 0{,}625 + (-100) \times (0{,}0625) = 256{,}57 \text{ kN}$$

1.b) Shear force at C



For unit vertical load

$\eta_A = 0 \quad \eta_{CA} = 10/16 = 0{,}625 \quad \eta_{CB} = -6/16 = -0{,}375 \quad \eta_B = 0$

$\eta_D = -3/16 = -0{,}1875$

$A_{\eta 1} = 0{,}5 \times 0{,}375 \times 6{,}0 = 1{,}125 \text{ m}$

$A_{\eta 2} = 0{,}5 \times 0{,}625 \times 10{,}0 = 3{,}125 \text{ m}$

$A_{\eta 3} = 0{,}5 \times 0{,}1875 \times 3{,}0 = 0{,}281 \text{ m}$

For unit moment

$\eta_A = \eta_D = \eta_B = \eta_D = 1/16 = 0{,}0625 \text{ m}^{-1}$

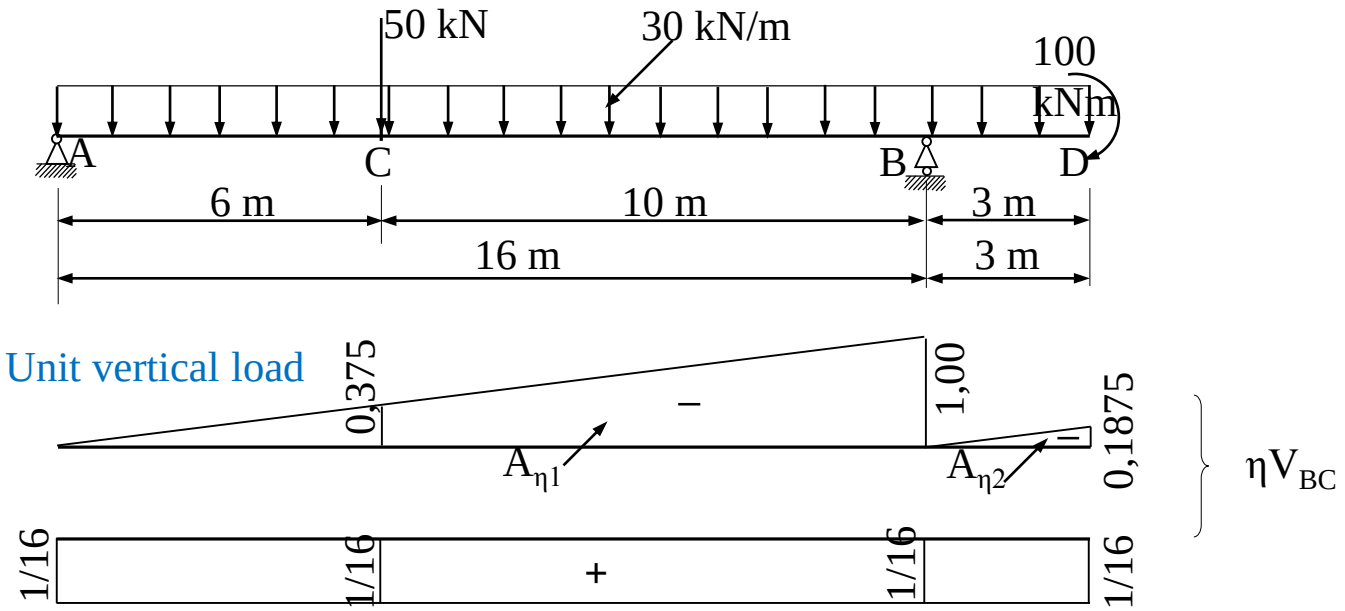
$V_{CA} = 30 \cdot (-1{,}125 + 3{,}125 - 0{,}281) + 50 \times 0{,}625 + (-100) \times (0{,}0625)$

$V_{CA} = 76{,}57 \text{ kN}$

$V_{CB} = 30 \cdot (-1{,}125 + 3{,}125 - 0{,}281) + 50 \times (-0{,}375) + (-100) \times (0{,}0625)$

$V_{CB} = 26{,}57 \text{ kN}$

1.c) Shear force at the left of B (V_{BC})



For unit vertical load

$$\eta_A = 0 \qquad \eta_{BD} = 0 \qquad \eta_D = -3/16 = -0{,}1875$$

$$\eta_C = -6/16 = -0{,}375 \qquad \eta_{BC} = -1$$

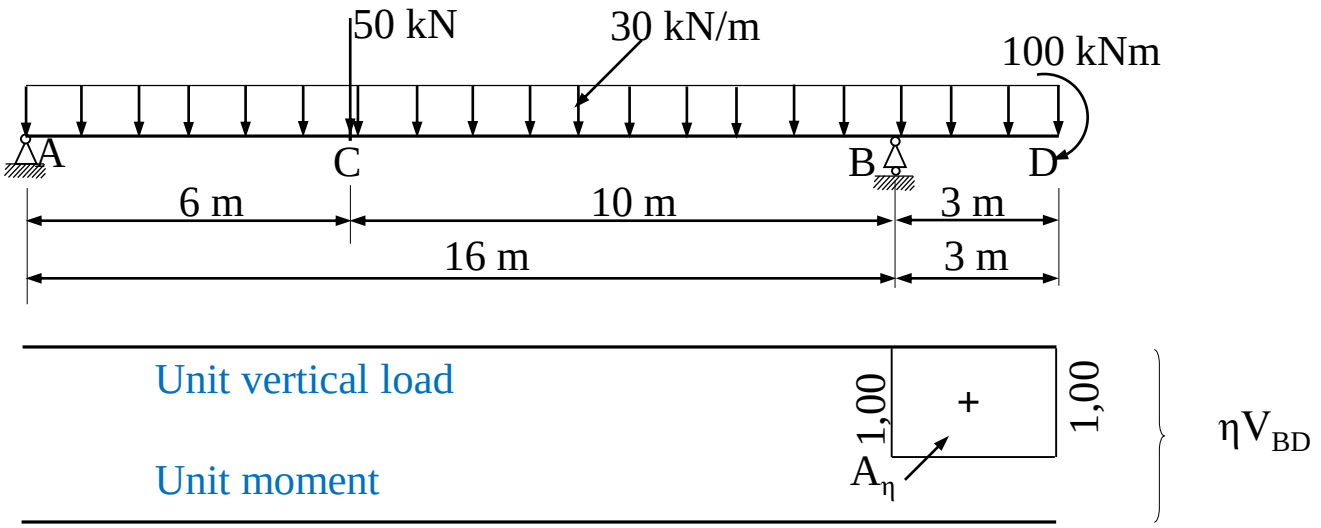
$$A_{\eta 1} = 0{,}5 \times 1{,}0 \times 16{,}0 = 8{,}00 \text{ m} \qquad A_{\eta 2} = 0{,}5 \times 0{,}1875 \times 3{,}0 = 0{,}281 \text{ m}$$

For unit moment

$$\eta_A = \eta_D = \eta_B = \eta_D = 1/16 = 0{,}0625 \text{ m}^{-1}$$

$$V_{BC} = 30 \cdot (-8{,}00 - 0{,}281) + 50 \times (-0{,}375) + (-100) \times (0{,}0625) = -273{,}43 \text{ kN}$$

1.d) Shear force at the rigth of the B (V_{BD})



For unit vertical load

$$\eta_A = \eta_C = \eta_{BA} = 0 \qquad \eta_{BD} = 1 \qquad \eta_D = 1 \text{ m}$$

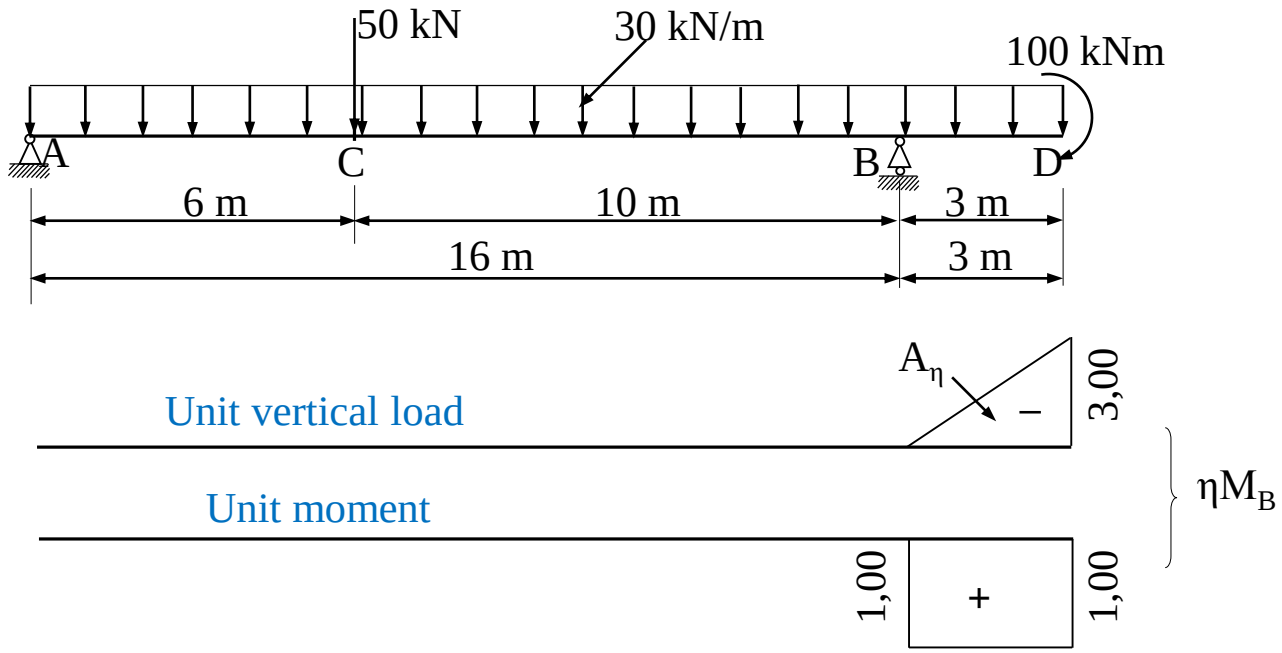
$$A_\eta = 0{,}5 \times 3{,}0 \times 1{,}0 = 1{,}5 \text{ m}$$

For unit moment

$$\eta_A = 0 \qquad \eta_B = \eta_D = 0$$

$$V_{BD} = 30 \times 1{,}50 = 45 \text{ kN}$$

1.e) Bending moment at B (M_B)



For unit vertical load

$\eta_A = 0 \qquad \eta_C = 0 \qquad \eta_{BA} = 0 \qquad \eta_B = 0 \qquad \eta_D = -3 \text{ m}$

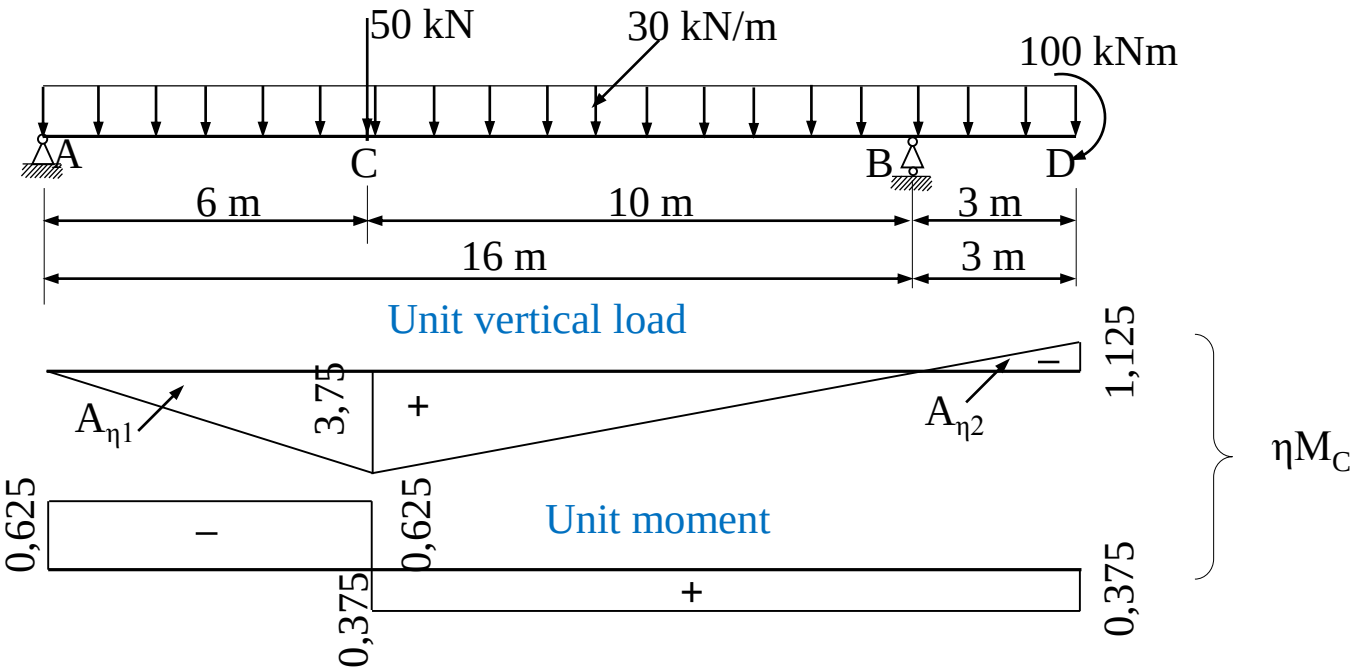
$A_\eta = 0{,}5 \times 3{,}0 \times 3{,}0 = 4{,}5 \text{ m}^2$

For unit moment

$\eta_A = \eta_D = 0 \qquad \eta_B = \eta_D = 1$

$M_B = 30 \times (-4{,}5) + 50 \times 0{,}0 + (-100) \times 1{,}0 = -235{,}00 \text{ kNm}$

1.f) Bending moment at C (M_C)



For unit vertical load

$\eta_A = 0 \qquad \eta_C = 6{,}0 \times 10{,}0 / 16{,}0 = 3{,}75 \text{ m} \qquad \eta_B = 0$

$\eta_D = -3{,}0 \times 6{,}0 / 16 = -1{,}125 \text{ m}$

$A_{\eta 1} = 0{,}5 \times 3{,}75 \times 16{,}0 = 30 \text{ m}^2 \qquad A_{\eta 2} = 0{,}5 \times 1{,}125 \times 3{,}0 = 1{,}6875 \text{ m}^2$

For unit moment

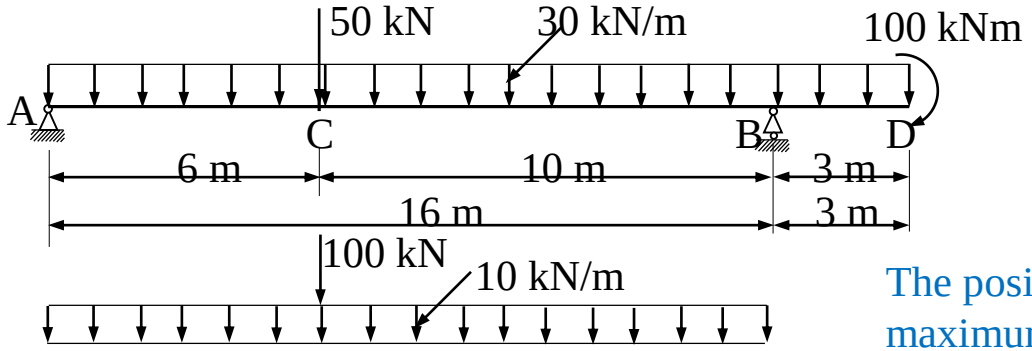
$\eta_A = -10/16 = -0{,}625 \qquad \eta_{CA} = 6/16 = 0{,}375$

$\eta_{CB} = -10/16 = -0{,}625 \qquad \eta_B = 6/16 = 0{,}375 \qquad \eta_D = 6/16 = 0{,}375$

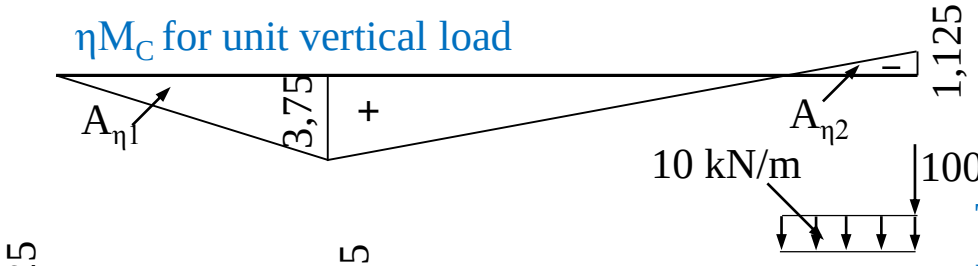
$M_{Cg} = 30 \cdot (30{,}00 - 1{,}6875) + 50 \times 3{,}75 + (-100) \times (0{,}375)$

$M_{Cg} = 999{,}375 \text{ kNm}$

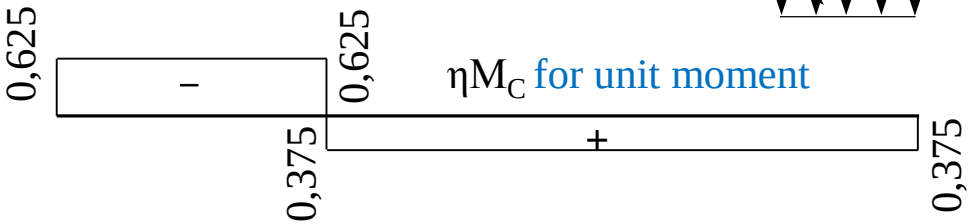
2) Maximum and minimum values of bending moment at C



The position that gives the maximum values of moment at C



The position that gives the minimumvalues of moment at C



$$M_{Cg} = 999{,}375 \text{ kNm}$$

Maximum bending moment

$$M_{Cq1} = 10 \times 30{,}00 + 100 \times 3{,}75 = 675{,}00 \text{ kNm}$$

$$M_{C,enb.} = 999{,}375 + 675{,}00 = 1674{,}375 \text{ kNm}$$

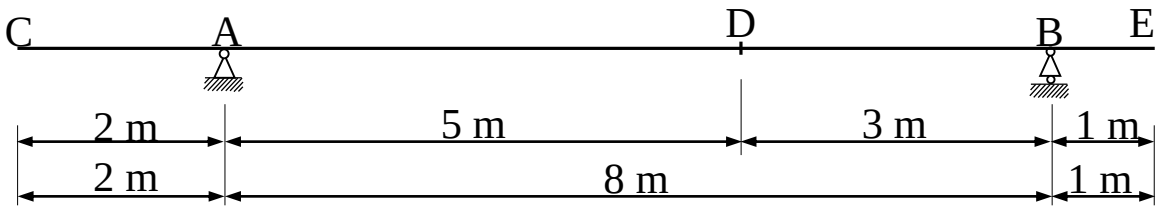
Minimum bending moment

$$M_{Cq2} = 10 \times (-1{,}6875) + 100 \times (-1{,}125) = -129{,}375 \text{ kNm}$$

$$M_{C,enk.} = 999{,}375 - 129{,}375 = 870{,}00 \text{ kNm}$$

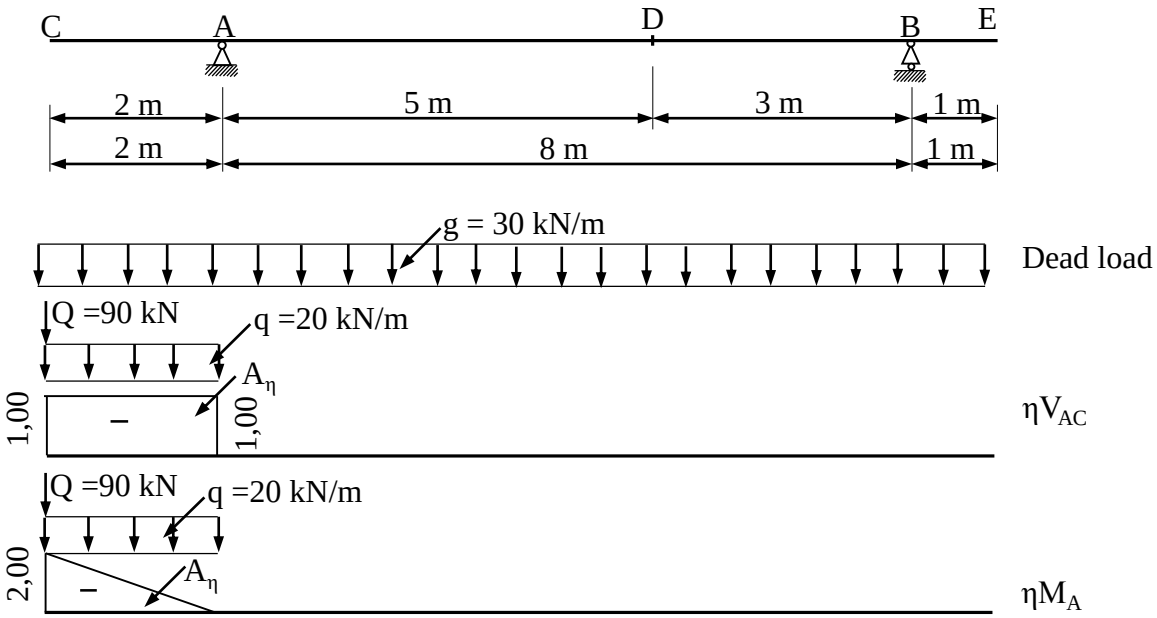
Example 3:

Estimate the maximum and minimum values of vertical reaction of support A, shear force and bending moment at A and D due to 30 kN/m uniform distributed dead load, 20 kN/m uniform distributed live load and 90 kN point live load.



Solution:

There are R_A support reaction at A, V_{AD} shear force at the right of A, V_{AC} shear force at the left of A, $M_{AC}=M_{AD}=M_A$ bending moment at A, V_D shear force and M_D bending moment at D.



a) V_{AC} shear force

$\eta_A = -1 \qquad \eta_C = -1 \qquad A_\eta = 1{,}0 \times 2{,}0 = 2{,}00 \ \text{m}$

Dead load $\qquad V_{AC,g} = 30 \times (-2{,}00) = -60 \ \text{kN}$

Live load $\qquad V_{AC,q1} = 20 \times (-2{,}00) + 90(-1) = -130 \ \text{kN}$

$V_{AC,enb.} = -60 - 130 = -190 \ \text{kN} \qquad V_{AC,enk.} = -60 + 0 = -60 \ \text{kN}$

b) M_A bending moment

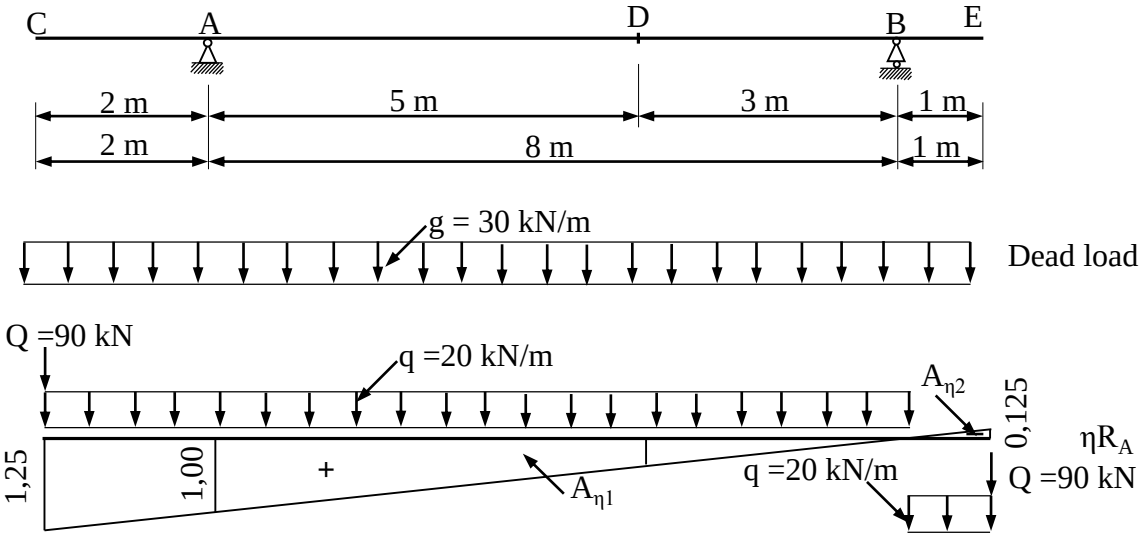
$\eta_A = 0 \qquad \eta_C = -2 \ \text{m} \qquad A_\eta = 0{,}5 \times 2{,}0 \times 2{,}0 = 2{,}00 \ \text{m}^2$

Dead load $\qquad M_{A,g} = 30 \times (-2{,}00) = -60 \ \text{kNm}$

Live load $\qquad M_{A,q1} = 20 \times (-2{,}00) + 90(-2) = -220 \ \text{kNm}$

$M_{A,enb.} = -60 - 220 = -280 \ \text{kNm} \qquad M_{A,enk.} = -60 + 0 = -60 \ \text{kNm}$

c) R_A support reaction



$$\eta_A = 1 \qquad \eta_C = 1 + 2/8 = 1{,}25 \qquad \eta_B = 0 \qquad \eta_E = -1/8 = -0{,}125$$

$$A_{\eta 1} = 0{,}5 \times 1{,}25 \times 10{,}0 = 6{,}25 \text{ m} \qquad A_{\eta 2} = 0{,}5 \times 0{,}125 \times 1{,}0 = 0{,}0625 \text{ m}$$

Dead load

$$R_{A,g} = 30 \times (6{,}25 - 0{,}0625) = 185{,}625 \text{ kN}$$

Live load

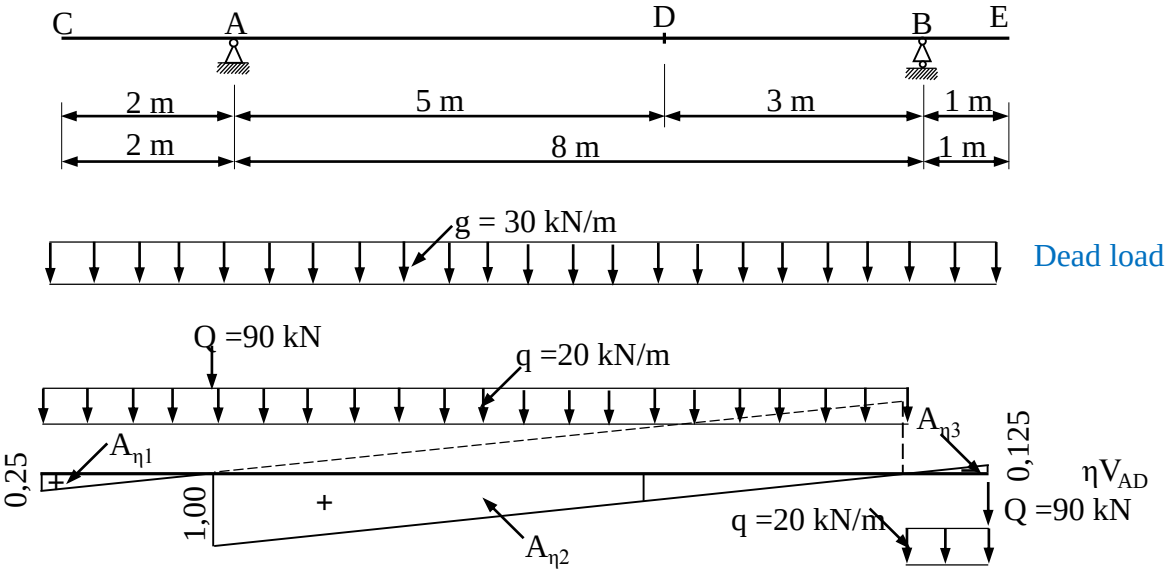
$$R_{A,q1} = 20 \times (6{,}25) + 90 \times 1{,}25 = 237{,}50 \text{ kN}$$

$$R_{A,q2} = 20 \times (-0{,}0625) + 90 \times (-0{,}125) = -12{,}50 \text{ kN}$$

$$R_{A,enb.} = 185{,}625 + 237{,}50 = 423{,}125 \text{ kN}$$

$$R_{A,enk.} = 185{,}625 - 12{,}50 = 173{,}125 \text{ kN}$$

d) V_{AD} shear force



$\eta_{AD} = 1 \qquad \eta_C = 2/8 = 0{,}25 \qquad \eta_{AC} = 0 \qquad \eta_B = 0$

$\eta_E = -1/8 = -0{,}125 \qquad A_{\eta 1} = 0{,}5 \times 0{,}25 \times 2{,}0 = 0{,}25 \text{ m}$

$A_{\eta 2} = 0{,}5 \times 1{,}0 \times 8{,}0 = 4{,}0 \text{ m} \qquad A_{\eta 3} = 0{,}5 \times 0{,}125 \times 1{,}0 = 0{,}0625 \text{ m}$

Dead load $\qquad V_{AD,g} = 30 \times (0{,}25 + 4{,}0 - 0{,}0625) = 125{,}625 \text{ kN}$

Live load

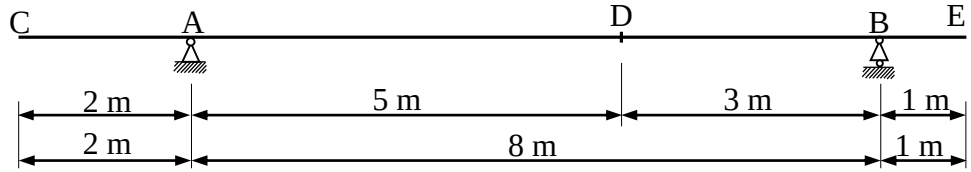
$V_{AD,q1} = 20 \times (0{,}25 + 4{,}00) + 90 \times 1{,}00 = 175{,}00 \text{ kN}$

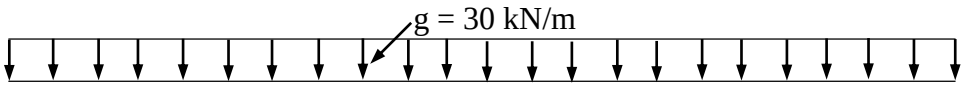
$V_{AD,q2} = 20 \times (-0{,}0625) + 90 \times (-0{,}125) = -12{,}50 \text{ kN}$

$V_{AD,enb.} = 125{,}625 + 175{,}00 = 300{,}625 \text{ kN}$

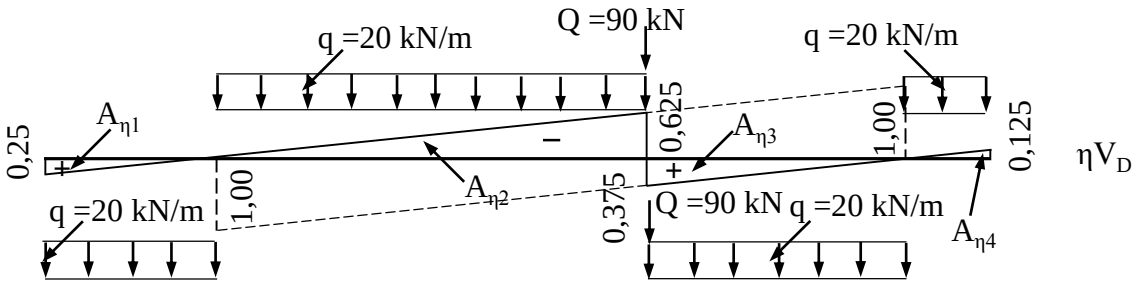
$V_{AD,enk.} = 125{,}625 - 12{,}50 = 113{,}125 \text{ kN}$

e) V_D shear force





Dead load



$\eta_C = 2/8 = 0{,}25 \quad \eta_A = 0 \quad \eta_{DA} = 3/8 = 0{,}375 \quad \eta_{DB} = -5/8 = -0{,}625$

$\eta_B = 0 \quad \eta_E = -1/8 = -0{,}125$

$A_{\eta 1} = 0{,}5 \times 0{,}25 \times 2{,}0 = 0{,}25 \text{ m} \qquad A_{\eta 4} = 0{,}5 \times 0{,}125 \times 1{,}0 = 0{,}0625 \text{ m}$

$A_{\eta 2} = 0{,}5 \times 0{,}625 \times 5{,}0 = 1{,}625 \text{ m} \qquad A_{\eta 3} = 0{,}5 \times 0{,}375 \times 3{,}0 = 0{,}5625 \text{ m}$

Dead load

$V_{D,g} = 30 \times (0{,}25 - 1{,}625 + 0{,}5625 - 0{,}0625) = -26{,}250 \text{ kN}$

Live load

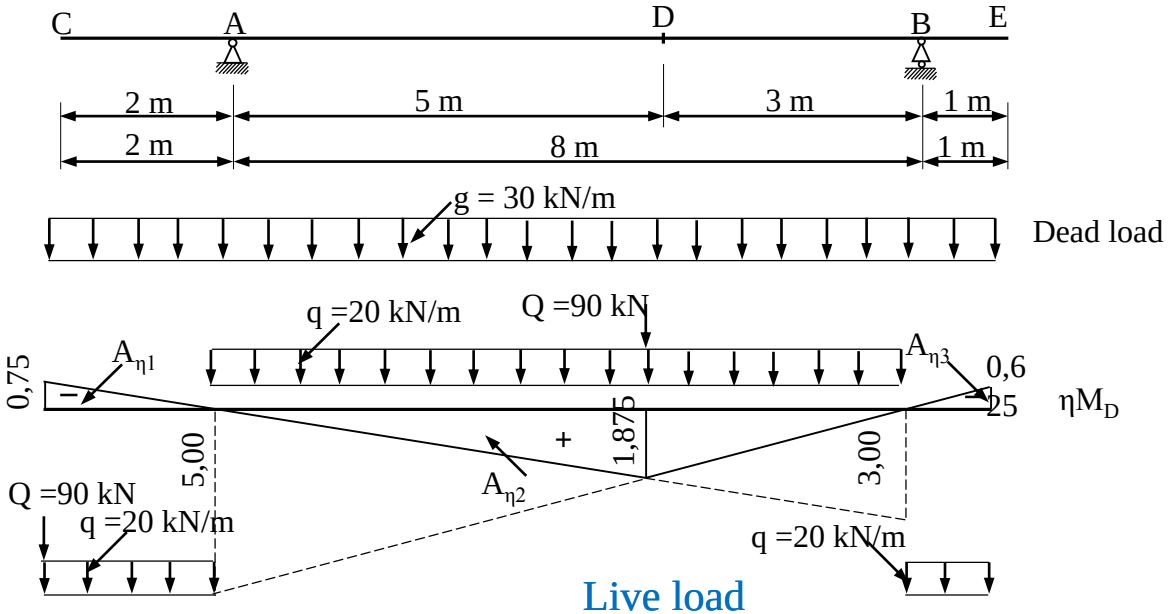
$V_{D,q1} = 20 \times (-1{,}625 - 0{,}0625) + 90 \times (-0{,}625) = -90{,}00 \text{ kN}$

$V_{D,q2} = 20 \times (0{,}25 + 0{,}5625) + 90 \times (0{,}375) = 50{,}00 \text{ kN}$

$V_{D,enb.} = -26{,}250 - 90{,}00 = -116{,}250 \text{ kN}$

$V_{D,enk.} = -26{,}250 + 50{,}00 = 23{,}750 \text{ kN}$

f) M_D bending moment



Live load

$\eta_C = -2{,}0 \times 3{,}0 / 8{,}0 = -0{,}75 \text{ m} \quad \eta_A = 0 \quad \eta_D = 3{,}0 \times 5{,}0 / 8 = 1{,}875 \text{ m}$

$\eta_B = 0 \quad \eta_E = -1{,}0 \times 5{,}0 / 8 = -0{,}625 \text{ m}$

$A_{\eta 1} = 0{,}5 \times 0{,}75 \times 2{,}0 = 0{,}75 \text{ m}^2 \quad A_{\eta 2} = 0{,}5 \times 1{,}875 \times 8{,}0 = 7{,}50 \text{ m}^2$

$A_{\eta 3} = 0{,}5 \times 0{,}625 \times 1{,}0 = 0{,}3125 \text{ m}^2$

Dead load

$M_{D,g} = 30 \times (-0{,}75 + 7{,}50 - 0{,}3125) = 193{,}125 \text{ kNm}$

Live load

$M_{D,q1} = 20 \times 7{,}50 + 90 \times 1{,}875 = 318{,}75 \text{ kNm}$

$M_{D,q2} = 20 \times (-0{,}75 - 0{,}3125) + 90 \times (-0{,}75) = -88{,}750 \text{ kNm}$

$M_{D,enb.} = 193{,}125 + 318{,}75 = 511{,}875 \text{ kNm}$

$M_{D,enk.} = 193{,}125 - 88{,}750 = 104{,}375 \text{ kNm}$

Reference:

- Prof. İbrahim EKİZ, YAPI STATİĞİ 1, BİRSEN YAYINEVİ

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